

Illinois State University – Physics MS Thesis Proposal

Two Perspectives on Plasma Transport: Defect Diffusion & Spectral Line Shapes

Levi Webb
lrwebb@ilstu.edu
webb45@llnl.gov
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Committee Chair: Dr. Matt Caplan
Committee Member 2: Dr. Neil Christensen
Committee Member 3: Dr. Q. Charles Su

Illinois State University, Department of Physics
Masters of Science in Physics

Background: Plasma

A plasma is a partially or fully ionized hot gas—often considered the fourth state of matter—which, under certain conditions of temperature and density, can behave not only as a fluid but also as a strongly coupled solid. They are found in a few places on Earth, such as lightning and the ionosphere. Additionally, stellar and compact objects are made of high energy density (HED) plasmas; however, the microscopic properties of these plasmas cannot be directly observed in astrophysical environments, nor can they be stably produced in a lab environment.

To address this challenge, I propose two computational approaches, both utilizing molecular dynamics (MD), to better understand the microphysics of HED plasmas. The first investigates the propagation of defects within cool, crystallized plasmas relevant to compact astrophysical objects. The second examines how particle dynamics in HED plasmas influence macroscopic opacities and spectral line shapes.

Plasmas are described by a few unique parameters. The first is a screening parameter,

$$\kappa = \frac{a}{\lambda_e}, \quad (1)$$

where $a = (4\pi n/3)^{-1/3}$ is the plasma ion spacing with number density n and λ_e is the electron screening length. Another convention, used by this work, is that $\kappa = n^{-1/3}/\lambda_e$. HED plasmas have small κ values (weak screening), so they are dominated by long-range Coulomb forces. This presents a computational challenge with modeling these intense environments, as it is non-trivial to determine where to truncate those long-range interactions.

Another important parameter is the Coulomb coupling parameter Γ , a ratio of potential to kinetic energy. In a strongly coupled one-component plasma (OCP), the system is wholly characterized by κ and the coupling parameter [4]

$$\Gamma = \frac{e^2 Z^2}{ak_B T}. \quad (2)$$

Strong coupling ($\Gamma \geq 1$) means that potential energy dominates, so a particle's motion is highly correlated with that of the particles around it and the system is highly dense. This explains how HED plasmas may behave as liquids and solids.

I will present my diffusion data with respect to a modified coupling parameter Γ^* , as introduced by Ref. [5] and represented by

$$\Gamma = \Gamma^* \frac{\Gamma_{crit} e^{\kappa/0.62}}{1 + \kappa/0.62 + 0.5 * (\kappa/0.62)^2}. \quad (3)$$

where $\Gamma_{crit} = 175.7$ is the coupling at which an unscreened ($\kappa = 0$) OCP crystallizes and $0.62 = an^{1/3}$ is the conversion factor between the κ in Eq. 1 and the dimensionless κ used in this proposal. This standardization constitutes a melting point such that the plasma behaves as a solid for $\Gamma^* \gtrsim 1$. Since $\Gamma \propto \frac{1}{T}$, I will conveniently refer to Γ as an analogue to inverse temperature throughout this proposal.

Defect Diffusion in Coulomb Crystals

In a strongly coupled HED plasma, such as those found in the crusts of neutron stars or cores of white dwarfs, Coulomb forces hold charged particles together in a solid crystal lattice. Within these crystals, thermal fluctuations occasionally lead to rare lattice defects wherein particles to move off of their sites and into the interstitial spaces, leaving behind vacancies. This formation mechanism means that defects always form in pairs.

These defects propagate throughout the crystal as quasi-particles. An interstitial particle might take the lattice site of another particle through a combination of its own repulsive force and the other particle’s thermal fluctuations. Then, the particle that was previously in an interstitial space is no longer a defect, and the defect has “moved” via the particle that has been displaced. Particles near a vacancy will also be pushed into the space by others around it, causing the vacancy to “move.” This defect motion can be tracked by measuring the mean-squared displacement (MSD) of particles from their initial positions. Since particles unaffected by defects have negligible displacement, the MSD serves as a valid probe of defect motion.

The time derivative of the MSD produces a diffusion coefficient, D^* . It has the associated equation

$$D^* = N(D_{vacancy}^* + D_{interstitial}^*), \quad (4)$$

where N is the number of defect pairs. This quantity is dubbed D^* as opposed to D because it is not a true diffusion coefficient; rather, it is a normalized proxy for the defect’s jump frequency [5]. From the equation, it is clear that crystals may experience periods of quiescence wherein $N = 0$ and therefore $D^* = 0$. This occurs when all defect pairs meet and annihilate (i.e. an interstitial moves into a vacancy and thus becomes realigned with the grid).

Despite being a microscopic event, diffusion within Coulomb crystals has potentially macroscopic effects in cooling compact objects. Previously, this motion has been poorly understood, and areas of crystallization in current stellar models are simply assumed to be non-evolving [5, 2]. Recent works have developed routines to calculate D^* in Coulomb crystals [3, 5], but the jump rates of an individual interstitial or vacancy have yet to be resolved sufficiently. This project aims to address that gap by running long, high-fidelity simulations to capture and analyze these defect dynamics.

Proposed Methods

My investigation into the diffusion of defects in Coulomb crystals will make use of ISU’s supercomputing resources. I will use Sandia National Laboratory’s 3D classical molecular dynamics code, Large-scale Atomic/Molecular Massively Parallel Simulator [LAMMPS, 13], to simulate and evolve a body-centered cubic (BCC) lattice of OCP particles in time.

Defect pairs will naturally form due to thermal fluctuations, as aforementioned, but LAMMPS allows the user to artificially create a defect through the insertion or removal of a single particle. This fundamentally changes the diffusion coefficient equation:

$$D^* = N(D_{vacancy}^* + D_{interstitial}^*) + D_{artificial}^*. \quad (5)$$

It is evident that, with an artificial defect, the system will never experience quiescence (i.e. $D = D_{artificial}^*$ when $N = 0$). Thus, the averaged D^* from LAMMPS gives the diffusion coefficient of the individual defects if the simulation has no others.

On a large lattice, (e.g. the 16k-particle simulations of Ref. [5]), there is a greater chance that defect pairs form and they are less likely to annihilate. Additionally, these large lattice simulations are computationally expensive. If, instead, I use a smaller lattice (e.g. 2k particles), the probability of defect pair creation greatly decreases, and any that spontaneously form are likely to annihilate quickly. The shorter run times allow for more simulations, thereby resulting in a greater body of data and enhancing the accuracy of the resulting diffusion relation (see Fig. 1).

Preliminary Results

Among many other input parameters, LAMMPS takes the coupling parameter Γ and screening parameter κ . I make a selection of Γ^* values, then use Eq. 3 to obtain the corresponding Γ values for each κ I simulate. For data presentation (e.g. Fig. 3), I recalculate Γ^* so that the scatter plot is aligned vertically and it is easier to visually separate trends for each κ value.

By default, LAMMPS outputs particle positions at specified time steps. One may also specify a number of other outputs, including a diffusion coefficient at each time step (see Fig. 2). It calculates D^* via the MSD slope method discussed in the scientific background of this proposal. Due to thermal fluctuations, D^* varies over time; I use the value at the end of each simulation, once it has stabilized, and plot this with respect to the κ and Γ values of the

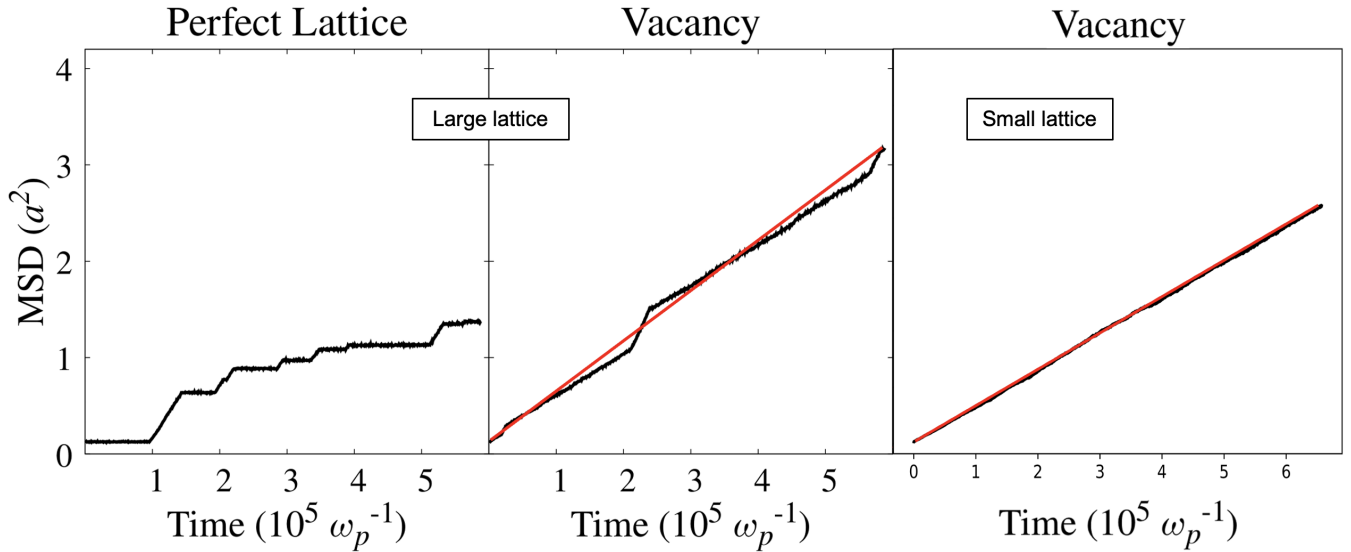


Figure 1: Plots of mean-squared displacement (MSD) over time (with reference to the plasma’s free electron oscillation frequency). The left & center plots show large lattice simulations (16k particles), whereas the right is a small lattice simulation (2k particles). In perfect lattice, there are periods of quiescence where the slope of the MSD is 0 (i.e. that $D^* = 0$). When there is an artificially inserted defect (in this case, a vacancy in the center and right plots) the system never experiences this. The red line averages the MSD for a simulation, the slope of which I use for a diffusion coefficient. This is the easiest way to obtain D^* . With this method, it is evident that a small simulation provides a D^* that is most accurate to the actual MSD slope.

system. Initially, I defined the onset of crystal freezing (where simulations are dominated by thermal noise) via the appearance of their MSD plots. These initial results are plotted in Fig. 7 in the appendix. However, heuristic analysis is insufficient for detailed molecular dynamics simulations.

To more rigorously define a diffusive system, I analyzed the LAMMPS particle position output files to track defect motion. By constructing displacement vectors and calculating the angle between successive time steps, I determined how frequently defects turn off their lattice columns for different values and combinations of κ and Γ (see App. A.B). I define a “turn” to be a change in direction exceeding 10° . If a defect turns more than a specified number of times (here, more than 100), the simulation is classified as diffusive. These values are empirically chosen, and will be further examined in the future to ensure that they are physically motivated.

The results in Fig. 3 show diffusion coefficients for both interstitials and vacancies over all simulated values of κ and their corresponding Γ^* values. The fainter, smaller points indicate that the corresponding simulation was non-diffusive, rather dominated by thermal noise. It is evident from these results (and fig. 7) that vacancies diffuse at lower temperatures than interstitials.

Vacancies diffuse more easily than interstitials because particles surrounding a vacancy are pushed into it by their neighbors, encouraging motion. In contrast, interstitials must exert force on a particle whose thermal motion is already directed away from the interstitial site, making their migration less likely. Since interstitials depend on thermal noise, vacancies diffuse at lower temperatures (evident in both Figs. 7 & 3, regardless of data analysis method.)

In future work, I will fit a regression to the D^* plots to identify a diffusion trend. I will also investigate whether using 100 10° turns is the best threshold for classifying diffusion by analyzing the distribution of angles and turns across different κ and Γ^* values. Further examination of defect displacement vector rotations may reveal how crystal structure influences diffusion, motivating simulations in lattice types beyond BCC. If these steps do not resolve the differences observed between the MSD-based and turn-based diffusion classifications, I will continue to refine and compare these methods to ensure robust classification of diffusion regimes.

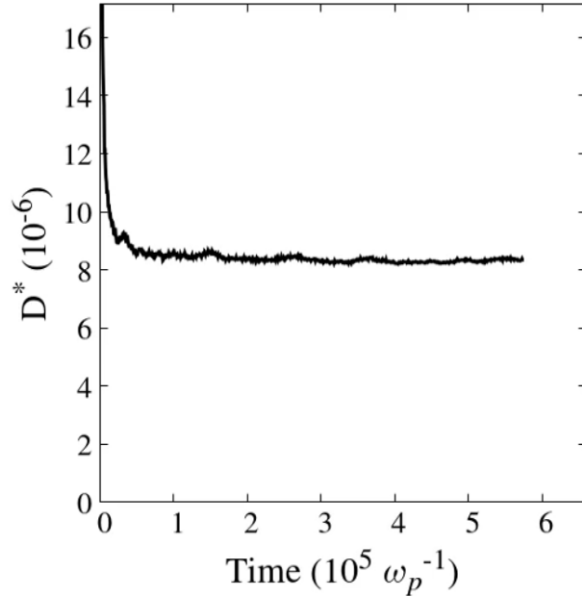


Figure 2: A plot of D^* over time for $\kappa = 0.5$, $\Gamma^* = 1$. This demonstrates that the value of D^* output by LAMMPS converges (i.e. the value taken from this plot would be $\approx 8 * 10^{-6}$).

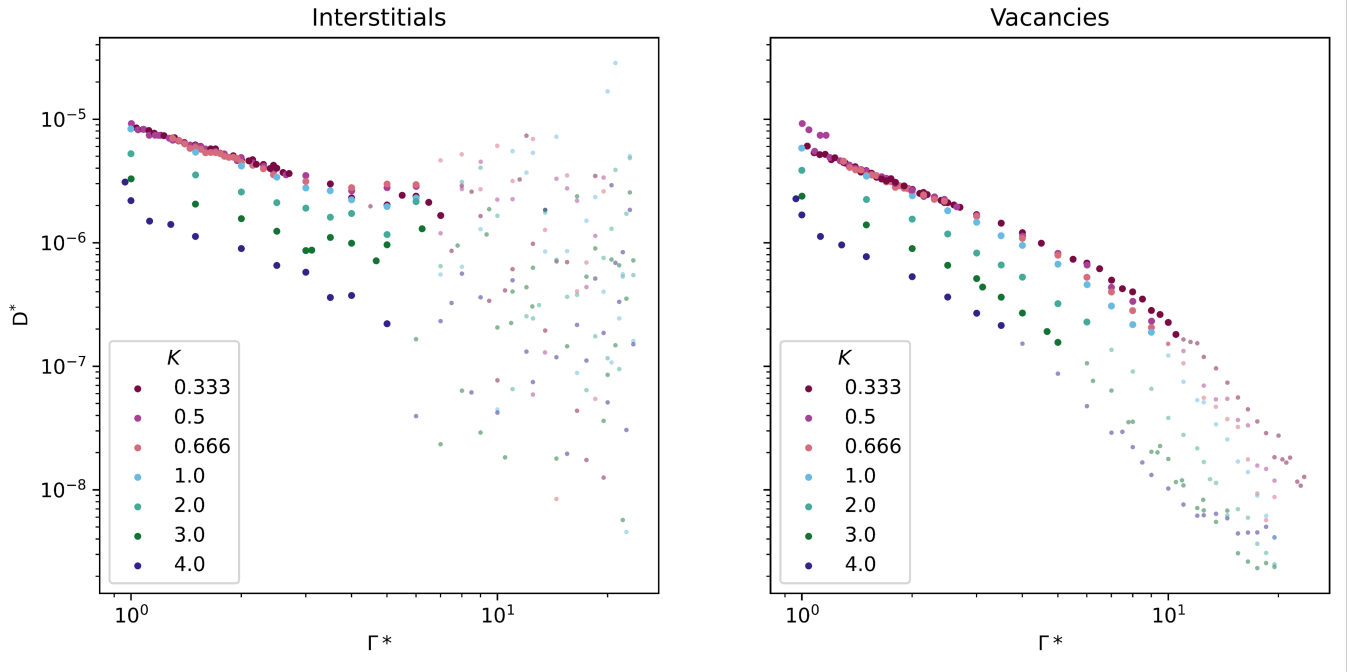


Figure 3: Diffusion coefficients for interstitials (left) and vacancies (right) with respect to κ and Γ^* . Each point represents a simulation; fainter, smaller points are non-diffusive.

Spectral Line Shape & Radiation Opacity

Many stellar properties (including temperature, main sequence lifespan, and the size and location of convective zones) are dependent upon or closely tied to the opacity (absorbance per unit density) of the body's constituent layers. Spectral line shapes in particular are important for determining temperature and density of observed celestial objects; understanding and accurately modeling them, then, is crucial in stellar astrophysics.

In compact objects, such as neutron stars and white dwarfs, plasma opacity has large implications for cooling rates. However, labs cannot measure opacities at these densities and temperatures, leading to uncertainties in stellar models [1]. Being that radiation is a major energy transport mechanism in stars and compact objects, their spectral line shapes offer opportunities to probe opacity conditions, and developing convincing MD models for observed spectra can inform and vastly improve stellar models.

Current line shape models do not agree with observations of plasmas in the HED regime. Most notably, there is a discrepancy in the observed and calculated line shapes of bound-bound and bound-free transitions in iron plasmas at high temperatures that is less prevalent in a cooler iron plasma (see Fig. 4). Previous studies [e.g. 10] have shown that variances in iron group opacities strongly affect stellar models due to their abundance in these bodies. Thus, discrepancies in opacity models greatly impact the accuracy of stellar models. In order to better model these opacities, it is important to include the microphysics of these plasmas.

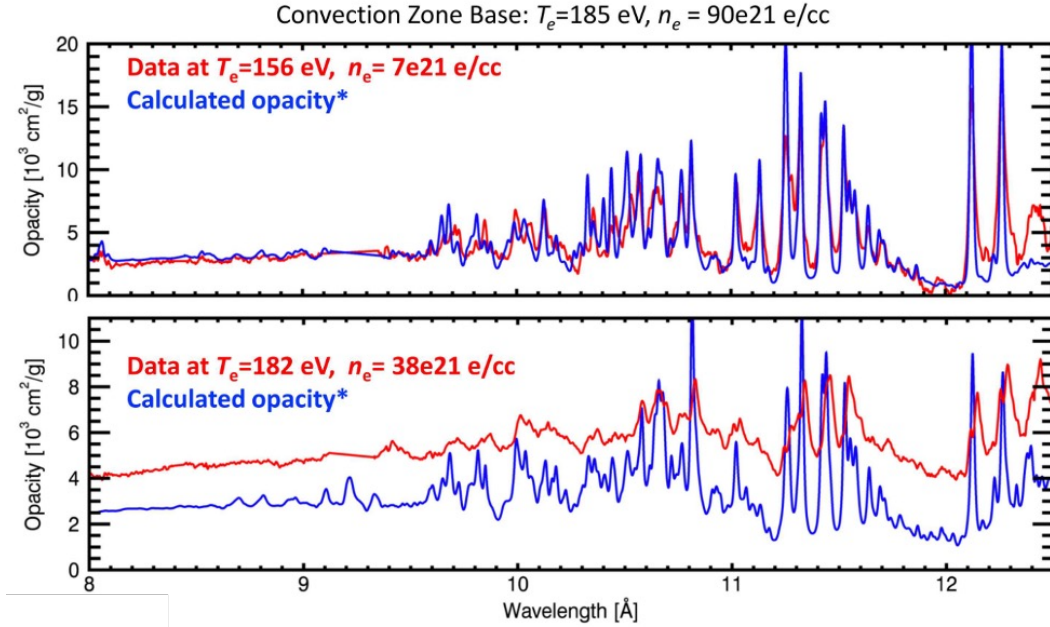


Figure 4: A plot of iron plasma opacities at two temperatures and electron number densities. There is a discrepancy between the observed and modeled line shapes of bound-free transitions (horizontal lines) and bound-bound (spectral lines) at higher temperatures and densities in iron plasmas [1].

Since these plasmas are strongly coupled, and hence highly dense, electric field fluctuations are incredibly stochastic due to the chaotic and close motion of particles. These local electric fields, or microfields, play a central role in shaping spectral lines through mechanisms such as Stark broadening [6], where the energy levels of atoms and ions are perturbed by the electric field fluctuations.

Microfields are not only important for line broadening but also for line merging (wherein lines merge into a quasi-continuum) and pressure ionization (fluctuations cause an ion lose its bound electron and merge with the continuum). Both of these effects alter the continuum opacity and the overall radiative properties of the plasma. In the context of stellar and objects, this affects the inferred physical conditions from spectroscopic observations and, ultimately, our understanding of stellar structure and evolution. Thus, incorporating physically motivated microfield distributions—obtained from MD simulations—of particles in an HED plasma is crucial for rectifying the observed spectral discrepancies and improving the fidelity of stellar models.

Proposed Methods & Preliminary Results

My research on spectral line shape and radiation opacity of HED plasmas will be conducted at Lawrence Livermore National Lab (LLNL) during a summer Defense Sciences and Technology Internship as an HED Science fellow in Strategic Deterrence. During this time, I will use LLNL’s massively parallel MD code, **ddcMD**, to model microfield distributions in HED plasmas. Accurate MD simulations are critical for accurate spectral line shape and opacity calculations.

ddcMD uses classical physics with a particle-particle-particle mesh (P^3M) algorithm for calculating electrostatic potentials, which are then used to calculate the microfields experienced by particles in the plasma. The P^3M method employs the truncation of interactions mentioned in the background of this project by calculating a direct sum for the potential of nearby particles, but modeling further particles as a grid for ease of calculating their potentials.

Current opacity models are outdated and semi-empirical [e.g. 9, 6]. Incorporating empirically derived microfield distributions from MD simulations will enable more accurate line shape calculations when integrated into codes such as the line broadening program **Xenomorph** [8]. This is an end goal of the project that will be achieved following the conclusion of my participation, but that I am still working toward. While **Xenomorph** currently uses semi-classical straight-line particle trajectories—suitable for weakly coupled, low-density plasmas—this approach breaks down in the HED, strongly coupled regime of interest.

My role in this project for the fiscal year 2025 is to calculate microfield distributions (preliminary results shown in Fig. 5), spanning from strong to weak coupling, for a range of temperatures, densities, simulation sizes, and other parameters to ensure that **ddcMD** has a physically-motivated and accurate microfield generator. From my results, **ddcMD** will be continually updated by the developer to prepare to merge with **Xenomorph**.

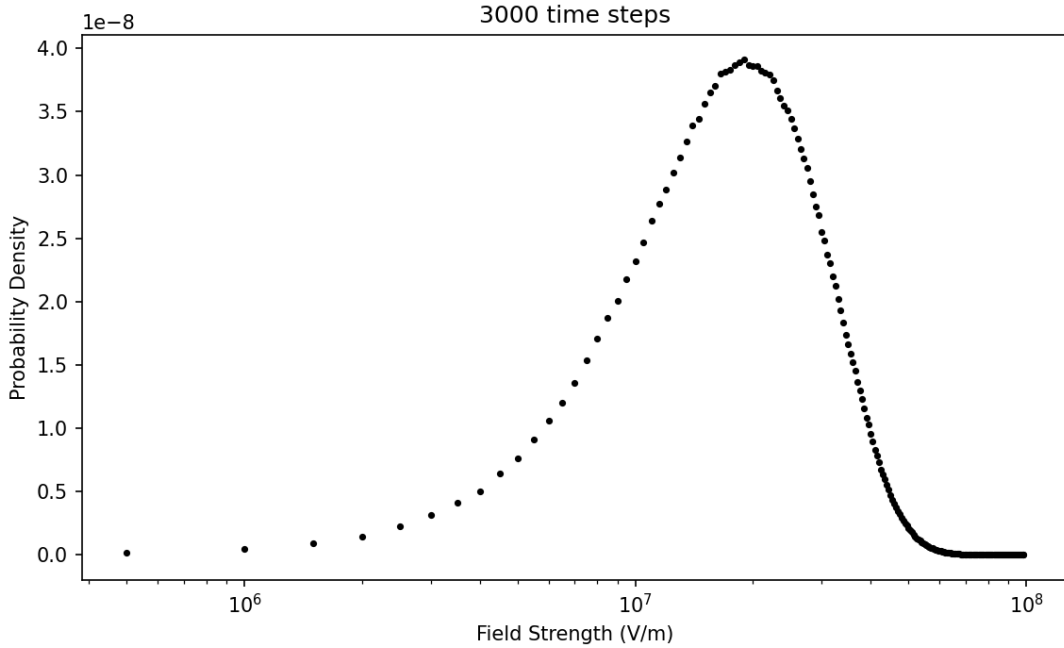


Figure 5: Preliminary microfield distribution of a 1keV hydrogen plasma. The simulation includes electrons and was run to 3000 time steps (1 time step is $5 * 10^{-4}$ fs); typical simulations will run much longer to achieve a smoother and more accurate distribution. These results demonstrate the capability of **ddcMD** to resolve microfield distributions at high temperatures.

My microfield calculations will begin with cool hydrogen plasmas (such as that of Fig. 5) and systematically extend toward higher temperatures, various coupling parameters, and species with larger Z -values. In addition to simulation work, I am contributing to the user documentation for **ddcMD**, providing feedback and detailed commentary to facilitate future development and user adoption.

Ultimately, this investigation will provide a foundation for improved opacity models in astrophysics and fusion research, enabling more accurate interpretation of spectroscopic data from both laboratory and astrophysical HED plasmas.

Timeline

- **Spring 2025:** Complete LAMMPS diffusion simulations and initial diffusion analysis
- **Feb 2025:** Present a poster on preliminary LAMMPS results at the American Physical Society (APS) Prairie Section Meeting
- **April 2025:** Present LAMMPS research as both a poster and a talk at University Research Symposium
- **Summer 2025:** Simulate HED plasmas for a range of temperatures and extract the particle electrical fields via ddcMD at LLNL
- **1 July 2025:** Defend thesis proposal
- **Aug 2025:** Present preliminary ddcMD results at LLNL Student Showcase
- **Fall 2025:** Begin writing thesis, finalize any analysis on LAMMPS microfield calculations and complete remaining ddcMD work on defect motion & diffusion classification
- **17-21 Nov 2025:** Present talk on microfield work and poster on Coulomb crystal work at APS Plasma Physics Division meeting
- **Late March - Early April 2026:** Defend thesis
- **April 2026:** Present ddcMD research at the ISU University Research Symposium, make necessary thesis edits from committee feedback
- **May 2026 or earlier:** Finalize and submit thesis
- **Summer 2026:** Continue work at LLNL

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Appendices

A Diffusion Classification

A.A MSD

Diffusion was classified either heuristically for interstitials (if MSD largely increases the entire time) or numerically for vacancies. For example, the interstitials in Fig. 6 were deemed non-diffusive above $\Gamma = 1164$, since $\Gamma = 1358$ experiences a significant decrease in MSD. Vacancy diffusion at the same κ value is so much lower on average that I deemed an MSD $> 15\text{fm}$ to be an acceptable cutoff. With this limit, the only diffusive vacancies in Fig. 6 are $\Gamma = 3007$ and $\Gamma = 3395$.

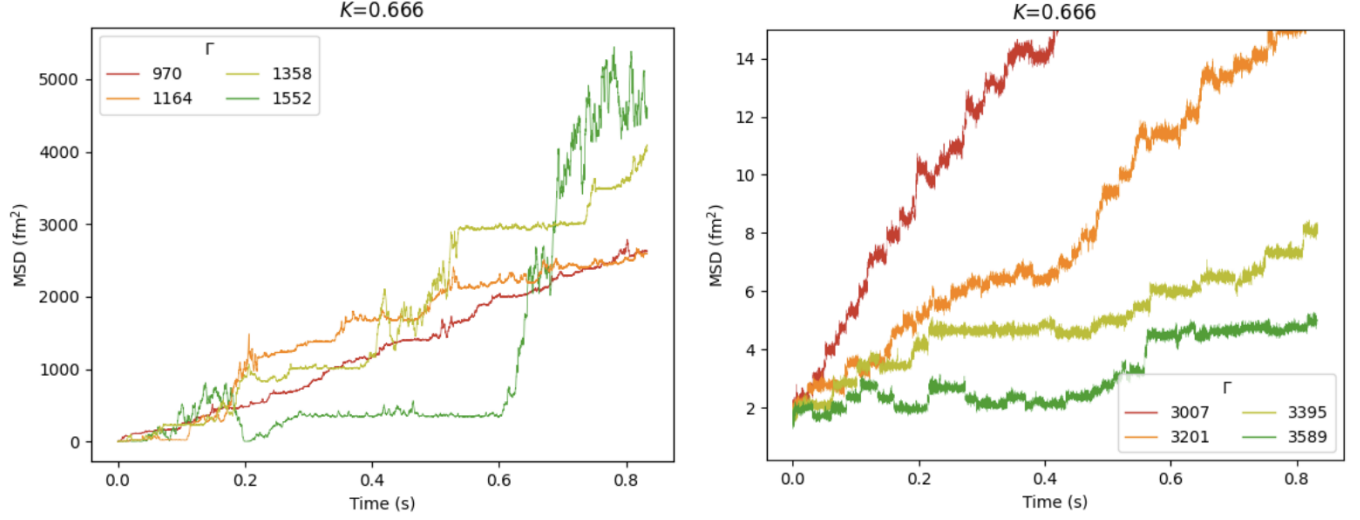


Figure 6: MSD over time for an interstitial (left) and vacancy (right). Note that the y-axes are not shared.

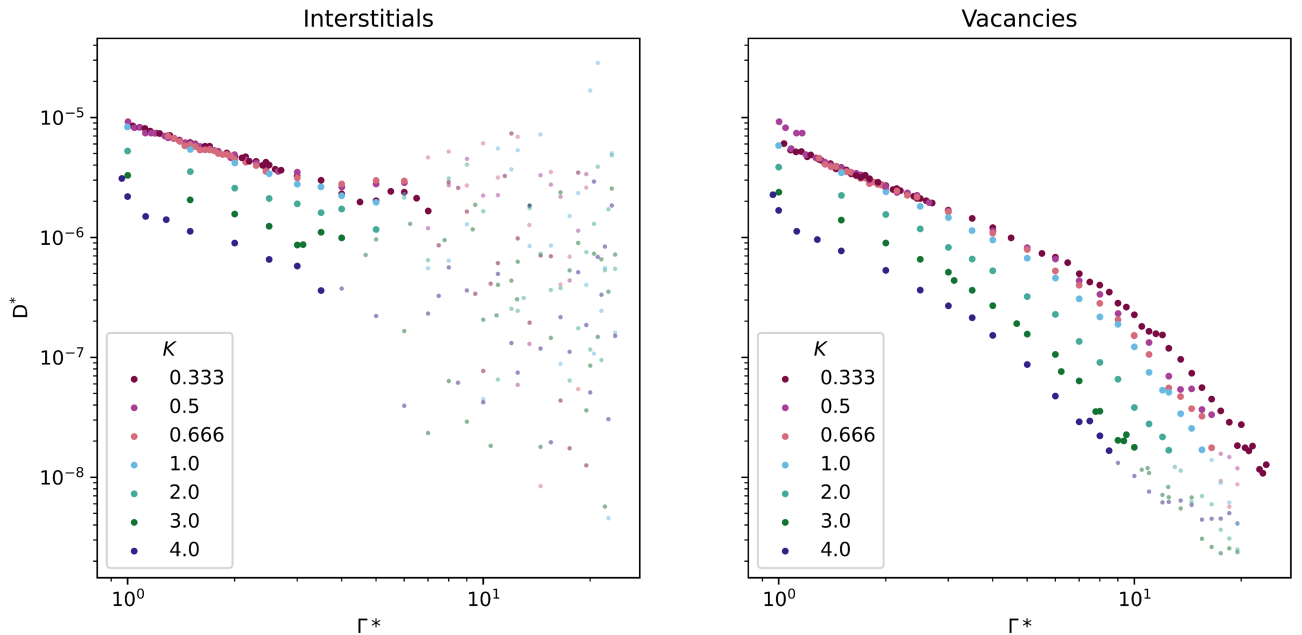


Figure 7: The same diffusion plot as Fig. 3, except with my initial MSD cutoff.

A.B Defect Turns

Tracking defects comes with some inherent uncertainty in how I define noise within a lattice spacing vs motion leaving a lattice. For example, if I define defect propagation as a movement larger than $0.25 \times (\text{lattice spacing})$, a particle that moves 30% of the way to its neighbor but then returns would still be counted as defect motion, even though the particle did not become a defect.

A defect with insufficient energy to turn off of its lattice site will experience a 1D random walk. In this case, the system is not truly diffusive, and I dim its diffusion coefficient on Fig. 3.

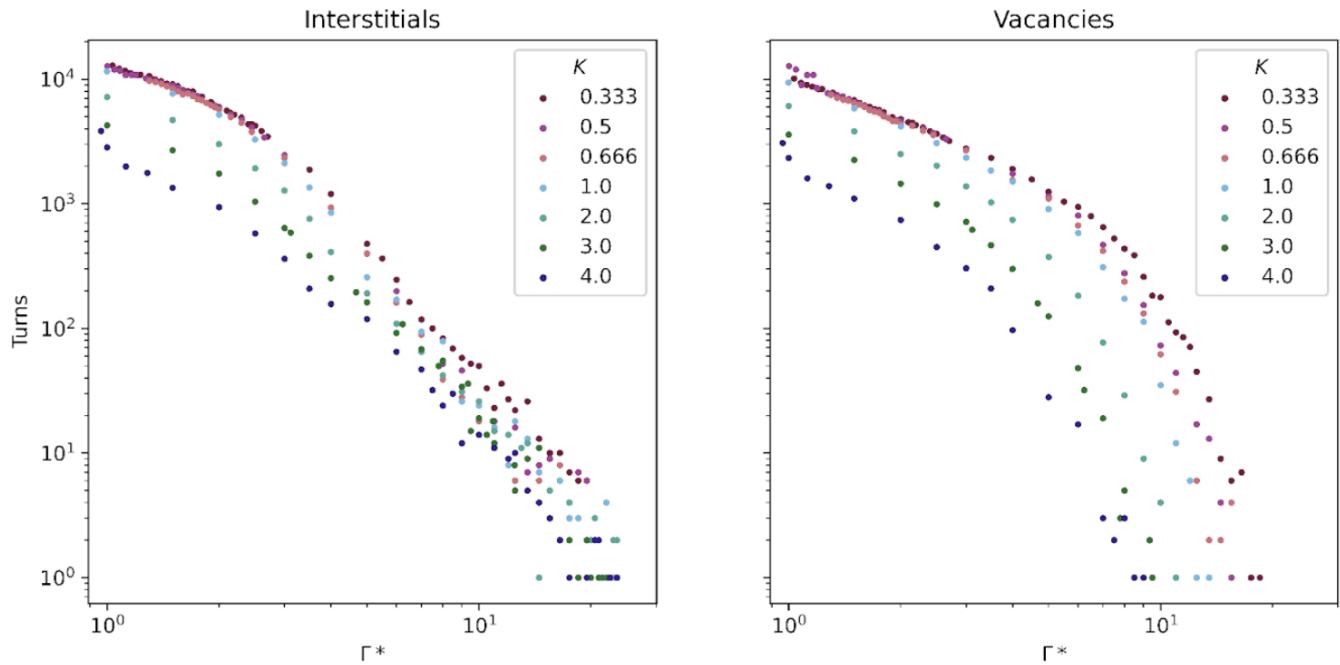


Figure 8: A plot showing turns vs Γ^* , where a turn is defined as a defect displacement vector angle $> 10^\circ$.